

1 Orthogonal projection in a Hilbert space

Let H be a Hilbert space and V a closed subspace of H . Note $\pi_V(f)$ is the orthogonal projection of f onto $V \in H$. Then

1. $\pi_V(f) \in V$;
2. $\langle f - \pi_V(f), g \rangle = 0$ for all $g \in V$

Show

1. For any $f \in H$, $\|f\|^2 = \|\pi_V(f)\|^2 + \|f - \pi_V(f)\|^2$

Since H is a Hilbert space, $f \in H$ can be decomposed as $f = f_V + f_\perp$.
Then

$$\begin{aligned}\|f\|^2 &= \langle f, f \rangle = \langle f_V + f_\perp, f_V + f_\perp \rangle = \\ &= \langle f_V, f_V \rangle + 2\langle f_V, f_\perp \rangle + \langle f_\perp, f_\perp \rangle = \\ &= \|f_V\|^2 + \|f - \pi_V(f)\|^2\end{aligned}$$

This follows since $f_\perp$ is the orthogonal complement of f_V .

2. For any $f, g \in H$, $\langle f, \pi_V(g) \rangle = \langle g, \pi_V(f) \rangle$

$$\begin{aligned}\langle f, \pi_V(g) \rangle &= \langle f_V + f_\perp, \pi_V(g) \rangle = \langle f_V, \pi_V(g) \rangle + \langle f_\perp, \pi_V(g) \rangle = \\ &= \langle \pi_V(f), \pi_V(g) \rangle \text{ since } f_V = \pi_V(f)\end{aligned}$$

In the same manner for g gives

$$\langle g, \pi_V(f) \rangle = \langle \pi_V(g), \pi_V(f) \rangle$$

Since the inner product is symmetric we can conclude that

$$\langle f, \pi_V(g) \rangle = \langle g, \pi_V(f) \rangle$$

3. Find an H and a subspace $V \subset H$ which is not closed in H and a $f \in H$ such that $\pi_V(f)$ does not exist.

Let $H = L^2[-1, 1]$ and V be the subspace of all continuous functions f on $[-1, 1]$

Define a function f' as

$$f'(x) = \begin{cases} 0, & x \in [-1, -\frac{1}{2}) \\ \frac{1}{2}, & x \in [-\frac{1}{2}, \frac{1}{2}] \\ 0, & x \in (\frac{1}{2}, 1] \end{cases}$$

Note that f' is the limit of a sequence of continuous functions, but is not

continuous and therefore is not contained in V . Therefore V is not closed. Now consider the orthogonal complement of f' , $f_{\perp} \in H$

$$f_{\perp} = \begin{cases} \frac{1}{2}, x \in [-1, -\frac{1}{2}) \\ 0, x \in [-\frac{1}{2}, \frac{1}{2}] \\ \frac{1}{2}, x \in (\frac{1}{2}, 1] \end{cases}$$

$f_{\perp} \in H$, but f' is not in V . Therefore for $f = f' + f_{\perp}$, $\pi_V(f)$ does not exist.

2 A reproducing kernel is positive definite

Let H be RKHS and $f : \mathcal{X} \mapsto \mathbb{R}$ by the kernel, K . This means

1. for any $y \in \mathcal{X}$, $K(\cdot, y) \in H$
2. for any $f \in H$, and $y \in \mathcal{X}$, $\langle f, K(\cdot, y) \rangle = f(y)$

1. Show that K is symmetric.

Since $K \in H$ choose $f = K(\cdot, y')$. Then by property 2.
 $\langle K(\cdot, y'), K(\cdot, y) \rangle = K(y', y)$

Likewise we also have
 $\langle K(\cdot, y), K(\cdot, y') \rangle = K(y, y')$

Since the inner product is symmetric $K(y', y) = K(y, y')$

2. Show that K is positive definite. Let x_i and $x_j \in \mathcal{X}$. Then for $\alpha_i \in \mathbb{R}$

$$\begin{aligned} \sum_i \sum_j \alpha_i \alpha_j K(x_i, x_j) &= \sum_i \sum_j \alpha_i \alpha_j \langle K(x_i, \cdot), K(x_j, \cdot) \rangle = \\ \langle \sum_i \alpha_i K(x_i, \cdot), \sum_j \alpha_j K(x_j, \cdot) \rangle &= \| \sum_i \alpha_i K(x_i, \cdot) \|^2 \geq 0 \end{aligned}$$

3. Show for any $f \in H$ and $x \in \mathcal{X}$, $|f(x)| \leq \|f\| \sqrt{K(x, x)}$

By the reproducing property of H , $f(x) = \langle f, K(\cdot, x) \rangle$. Then by the Cauchy-Schwarz inequality

$$\begin{aligned} |f(x)|^2 &= |\langle f, K(\cdot, x) \rangle|^2 \leq \|f\|^2 \|K(\cdot, x)\|^2 = \\ \|f\|^2 \langle K(\cdot, x), K(\cdot, x) \rangle &= \|f\|^2 K(x, x) \implies \\ |f(x)| &\leq \|f\| \sqrt{K(x, x)} \end{aligned}$$